Venkata Lakshmi Parimala Pasupuleti

Software Engineer · PropTech & Apartment AI · Python Backend Systems · Scalable Architecture · AI/ML · Startup Ownership

+1-703-261-5410 | lakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

PROFESSIONAL SUMMARY

Python software engineer with 9+ years building scalable backend systems and production AI in apartment and affordable housing software - the exact domain EliseAI operates in. At Entrata, I owned the backend systems that property managers, leasing agents, and compliance teams relied on daily across 100K+ housing units: scalable async processing, event-driven microservices, REST APIs, and the full infrastructure to run them reliably in production. I move fast and take full ownership - LeaseI is the proof point: a Python agentic AI platform I built concept-to-production solo in weeks, with full testing, observability, and CI integration. AgentProbe extends that: an open-source Python LLM evaluation platform with multi-provider routing, CI-integrated regression gates, and zero regressions shipped across 1,240+ test cases. The AI and ML side is where I bring something most engineers at EliseAI's stage won't: production-grade autonomous systems that already solve apartment problems. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Backend Systems & Infrastructure: Python (9yr), FastAPI, REST APIs, PostgreSQL, MySQL, Oracle, Redis, RabbitMQ (event-driven async), Docker (microservices), AWS (S3, EC2, RDS, Glue), CI/CD (GitHub Actions), Distributed Systems, Multi-Tenant SaaS, System Design, Scalable Architecture, Linux, Git

AI & Machine Learning: LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, LLM Evaluation, LLM-as-Judge, Hallucination Detection, RAG Pipelines, ChromaDB, Embeddings, Prompt Engineering, Anthropic API, OpenAI API, Claude Code, MCP, HITL, Guardrails, OCR, BERT, spaCy, NLP, PyTorch, scikit-learn

Reliability & Delivery: Testing Strategy (Unit, Integration, End-to-End), CI-Integrated Regression Gates, Observability & Monitoring, Performance Engineering, Production Debugging, Data Governance, PHI Compliance, Cross-Functional Stakeholder Delivery, Agile/Scrum, TypeScript, React.js, JavaScript

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - End-to-End Delivery

- Scoped and shipped a production-grade agentic AI system (LangGraph, FastAPI, Python) from ambiguous customer requirements to deployment - owned architecture, testing, observability, and stakeholder handoff end-to-end with full autonomy; used Claude Code to compress the build cycle and deliver working software in weeks.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn) on WHO and UK Road Safety datasets - $R^2 = 0.977$; designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and system design.

Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Backend systems for apartment operators at scale - Python, distributed architecture, event-driven processing** - owned the HUD income-limit compliance pipeline for 7 years (federal data → 180,000 domain records powering eligibility decisions for 100K+ housing units); RabbitMQ async event-driven processing with retry logic, idempotency guarantees, and dead-letter handling; Docker microservices (50% scalability improvement); 20+ production modules shipped across leasing, compliance, and document workflows with full ownership.
- **Reliable infrastructure built to run autonomously in production** - AWS S3/EC2/RDS/Glue; Redis caching; REST API design across multi-tenant SaaS; 200+ SQL query optimizations (30% performance gain); 99.5% uptime; 100% PHI compliance with Libsodium-encrypted pipelines (zero security incidents 7yr); built internal debugging tools reducing triage time by 40%.
- **Shipped with startup urgency across 120+ Agile sprints** - GitHub Actions CI/CD; sub-10 bug count 6+ consecutive months; 300+ peer code reviews; leveraged Claude Code and AI-assisted development to move faster; mentored 5+ engineers; worked directly with product, compliance, and operations stakeholders end-to-end.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) for apartment management and affordable housing compliance; data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

Leasel: Agentic AI Platform for Apartment Compliance - Concept to Production

PropTech · Python · Autonomous Systems · End-to-End

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Three-surface production-grade Python AI platform built solo from discovery through deployment - document intelligence (OCR, section-aware chunking, 100% citation correctness), 8-agent LangGraph support investigation (autonomous tool routing, memory, HITL escalation), and tenant self-service with deterministic guardrails. Full testing strategy, observability, and CI integration - 93.6% precision, 87.9% F1 across 1,240 apartment compliance test cases. Apartment AI built by someone who knows the domain.

AgentProbe: LLM Evaluation with CI-Integrated Regression Gate

Python · ML · Observability · Open Source

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Anthropic · OpenAI · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production Python LLM evaluation platform built end-to-end: 15-endpoint FastAPI service, 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate blocking regressions before they ship, multi-provider routing for cost optimization. Zero regressions shipped across 1,240+ test cases. Open-source.

AI on FHIR: Full-Stack NLP System - Customer-Facing Production

Full-Stack · APIs · Customer-Facing

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · LLMs · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack customer-facing AI application (Python backend + React.js/TypeScript frontend) translating natural-language queries into FHIR REST API calls via BERT embeddings and LLM inference - reduced retrieval time ~60%; HIPAA-compliant RBAC with authentication, authorization, and full data isolation.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017